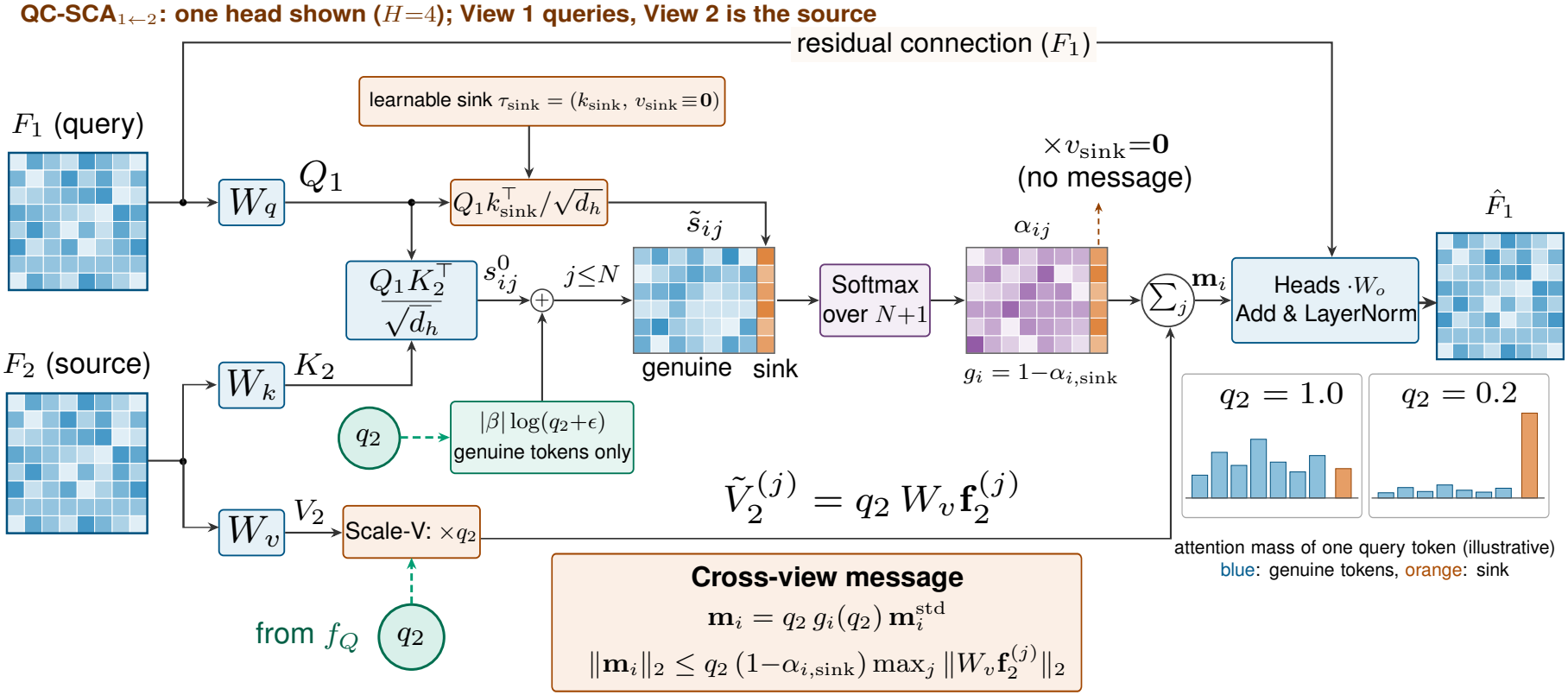

QC-SCA$_{1\leftarrow 2}$: one head shown ($H=4$); View 1 queries, View 2 is the source
residual connection (F_1)
F_1 (query)
learnable sink $\tau_{\text{sink}} = (k_{\text{sink}}, v_{\text{sink}} \equiv \mathbf{0})$
$\times v_{\text{sink}} = \mathbf{0}$
(no message)
$\hat{F}_1$
W_q
Q_1
$Q_1 k_{\text{sink}}^{\top}/\sqrt{d_h}$
$\tilde{s}_{ij}$
α_{ij}
$\frac{Q_1 K_2^{\top}}{\sqrt{d_h}}$
s^0_{ij}
$j \le N$
Softmax over $N+1$
$\sum_j$
$\mathbf{m}_i$
Heads $\cdot W_o$
Add & LayerNorm
genuine
sink
$g_i = 1-\alpha_{i,\text{sink}}$
F_2 (source)
W_k
K_2
q_2
$|\beta| \log(q_2+\epsilon)$
genuine tokens only
$q_2 = 1.0$
$q_2 = 0.2$
W_v
V_2
Scale-V: $\times q_2$
$\tilde{V}_2^{(j)} = q_2\, W_v \mathbf{f}_2^{(j)}$
attention mass of one query token (illustrative)
blue: genuine tokens, orange: sink
from f_Q
q_2
Cross-view message
$\mathbf{m}_i = q_2\, g_i(q_2)\, \mathbf{m}_i^{\text{std}}$
$\|\mathbf{m}_i\|_2 \le q_2\,(1-\alpha_{i,\text{sink}})\, \max_j \|W_v \mathbf{f}_2^{(j)}\|_2$